Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for: **Anaesthesia for Endoscopy on the Northern Site**

SUMMARY

This clinical guideline for the provision of anaesthesia in the Endoscopy suite on the Northern Site aims to support clinicians in providing safe anaesthetic care in endoscopy whilst also identifying those patients unsuitable for care in the endoscopy unit and providing an alternative pathway

KEY POINTS

- All patients referred for anaesthesia controlled sedation (ACS) in endoscopy will be reviewed by the endoscopy pre assessment service, supported if necessary, by the anaesthetic high risk clinic.
- This document provides clear acceptance criteria for patients requiring anaesthesia controlled sedation (ACS) in endoscopy.
- Patients deemed unfit for ACS in endoscopy will be booked on to a theatre list according to the pathway described.
- Patients known to require general anaesthesia must be booked on to a main theatre list.
- Patients should not be cancelled on the day of procedure without discussion with the anaesthetic consultant on call and a clear plan outlined and documented for their ongoing care.
- All endoscopy lists will adhere to the NatSSIPs 2 standards
- Equipment availability in endoscopy will comply with RCoA GPAS guidance on ‘Anaesthesia in the non-theatre environment’

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1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 This guideline builds on recent guidelines and a joint statement from the British Society of Gastroenterology Joint Advisory Group (JAG) and the Royal College of Anaesthetists (RCoA)[1,2 & 3] to provide up to date, locally relevant guidance on the provision of anaesthesia controlled sedation in the endoscopy unit.

2. BACKGROUND

2.1 There is increasing patient demand for anaesthesia controlled sedation (ACS) in the endoscopy unit. The endoscopy unit is a remote site and providing sedation for comorbid patients can be challenging. This document aims to support the anaesthetic team in identifying high risk patients and implementing appropriate plans for their ongoing management, either within the endoscopy suite or within the main theatre suite. The document will set standards of care for the endoscopy unit and provide guidance in the event of an emergency.

2.2 Patients known to require general anaesthesia must be booked onto a main theatre list (see appendix A)

2.3 In certain emergency scenarios patients undergoing anaesthesia controlled sedation will require conversion to general anaesthesia or advanced airway management. Endoscopy is fully equipped for this scenario

3. DEFINITIONS

3.1 Sedation levels

	Minimal Sedation Anxiolysis	Moderate Sedation/ Analgesia ("Conscious Sedation")	Deep Sedation/ Analgesia	General Anaesthesia
Responsiveness	Normal response to verbal stimulation	Purposeful** response to verbal or tactile stimulation	Purposeful** response following repeated or painful stimulation	Unarousable even with painful stimulus
Airway	Unaffected	No intervention required	Intervention may be required	Intervention often required
Spontaneous Ventilation	Unaffected	Adequate	May be inadequate	Frequently inadequate
Cardiovascular Function	Unaffected	Usually maintained	Usually maintained	May be impaired

Abbreviations Used in this Document

ACS: Anaesthesia controlled sedation

WHO: World Health Organisation

OGD: Oesophagogastroscopy

TCI: Target Controlled Infusions

OSA: Obstructive Sleep Apnoea

PACU: Post Anaesthesia care unit

HFNO: High flow nasal oxygenation

4. PRE OPERATIVE ASSESSMENT AND PREPARATION

4.1 Preassessment

- Prior to their procedure all patients will be pre assessed by the endoscopy clinic. This clinic is independent of the pre-operative assessment service provided by the anaesthesia department.
- Baseline investigations will be requested by the endoscopy pre assessment nursing team using locally agreed guidelines. https://royaldevonstaff.nhs.uk/download.cfm?ver=1760
- Patients with complex medical problems as defined by local pre assessment guidelines may require review in the anaesthetic high risk clinic by a consultant anaesthetist. https://royaldevonstaff.nhs.uk/download.cfm?ver=1760
- Investigations requested by both the endoscopy pre assessment team and the anaesthetic high risk clinic will be followed up by the endoscopy pre assessment team and referred back to the anaesthetic high risk clinic if necessary.
- All patients will receive the RCoA patient information leaflet on sedation prior to the day of their procedure. https://www.rcoa.ac.uk/sites/default/files/documents/2019-10/12-SedationExplainedweb.pdf

4.2 Acceptance Criteria for ACS in Endoscopy

The following criteria have been drawn up by the perioperative team (preoperative consultant anaesthetists/ preoperative nurses/ endoscopists) in order to facilitate a safe and consistent approach to the scheduling of patents for ACS/ anaesthesia in Endoscopy. They do not supersede clinical judgment.

General Anaesthesia

- Patients known to require general anaesthesia with advanced airway management are not accepted. They must be booked on to a theatre list (see Appendix A)

Paediatric patients

- Age <16 are not accepted
- Age 16-18 accepted by pre assessment consultant approval only

ASA grade

- ASA 1 or 2 or stable ASA 3 accepted
- 'Unstable ASA 3' accepted by pre assessment consultant approval only

Body Mass Index (BMI)

- BMI < 40 accepted
- BMI > 40 with co morbidity – accepted by pre assessment consultant approval only
- BMI >50 must be seen in the high risk anaesthetic clinic and booked on theatre list (Appendix A)

Obstructive Sleep Apnoea (OSA)

- Well controlled OSA with BMI <40 is accepted. Patient must bring their CPAP mask for post procedure care.
- OSA with BMI >40 is not accepted. Book onto theatre list (Appendix A)
- Patients with clinically suspected but undiagnosed OSA must not have their endoscopy deferred if it is clinically urgent.

Difficult airway

- Patients with known or suspected difficult airways should not be managed in endoscopy and should be booked on a theatre list (Appendix A)

Severe dementia/ learning difficulties/ autistic spectrum disorders

- Accepted by pre assessment consultant approval only
- If a side room is required book on a theatre list (Appendix A)
- Extra time may be required to facilitate reasonable adjustments. For example Ametop or premedication.

Neurological and neuromuscular disease

- Refer to the anaesthetic high risk clinic according to local pre op assessment guidance.
  https://royaldevonstaff.nhs.uk/download.cfm?ver=1760

- Patients with severe disease will be accepted with pre assessment consultant approval only
- See Appendix B for detailed criteria regarding PEG or RIG insertion in patients with known neuromuscular disease.

Cardiac disease

- Stable ASA 2 and 3 patients are accepted
- Unstable ASA 3 patients are accepted by pre assessment consultant approval only
- Patients that require invasive monitoring prior to sedation are not accepted and require booking on a theatre list (Appendix A)

Other anaesthetic issues

- Patients with known difficult IV access will require extra time
- Patients with known malignant hyperpyrexia (MH) will require extra time to prepare anaesthetic equipment.

4.3 Pathway for Patients Unsuitable for ACS in Endoscopy

For those patients who are deemed unsuitable for ACS in endoscopy, the following process should be followed:

- Confirm with the referring team (or gastro team if a GP referral) the required investigation and urgency
- If the patient requires colonoscopy then refer to the colorectal team via the "hot week' consultant. They will liaise with the referrer re booking on an elective theatre list (see Appendix A)
- If the patient requires an upper GI endoscopy then refer to the upper gastrointestinal team via the "hot week" consultant. They will liaise with the referrer re booking on an elective theatre list (see Appendix A)
- Theatre list bookings should be made via a SUR 1 form. Patients should not be booked on to the CEPOD list.

4.4 Cancellation on day of procedure

- Patients must not be cancelled on the day of the procedure without discussion with the consultant anaesthetist on call.
- Patients that are cancelled must be relisted for main theatres and have a documented timescale for their endoscopy conformed with the endoscopist on call.
- Patients who require investigation or optimization prior to their procedure must be urgently referred back to the anaesthetic high risk clinic with a plan for ongoing care.

5. PERIOPERATIVE MANAGEMENT

5.1 Fasting Guidelines

- Patients booked on the ACS list must be fasted for solids for 6 hours. As per current fasting guidance, they can drink clear fluids up to the time of endoscopy.
  https://royaldevonstaff.nhs.uk/download.cfm?ver=1782

5.2 Diabetic Patients

- Diabetic patients follow hospital guidelines and preferably placed first on the endoscopy list for the day.
  https://royaldevonstaff.nhs.uk/download.cfm?ver=11977

- Diabetic patients should not have an urgent procedure delayed to optimize their diabetic control.

5.3 Consent

- All patients should have received the RCoA leaflet “sedation” prior to the day of their procedure.
  https://www.rcoa.ac.uk/sites/default/files/documents/2019-10/12-SedationExplainedweb.pdf
- Verbal consent should be sought and documented for ACS on the day of procedure according to AAGBI guidance (16) This should include potential for awareness under sedation and conversion to general anaesthesia.
  https://anaesthetists.org/Portals/0/PDFs/Guidelines PDFs/Guideline_consent_for_anaesthesia_2017_final.pdf?ver=2018-07-11-163753-600&ver=2018-07-11-163753-600

5.4 Minimal Staffing for Theatre and Recovery

At least one of the endoscopy team should have completed ILS. All staff should have passed their BLS mandatory training.

- Endoscopist
- Endoscopy Nurses or 1 Endoscopy Nurse and a Health Care Worker
- Anaesthetist
- Operating Department Practitioner / Anaesthetic Nurse
- Recovery Nurse

5.5 Briefing & Checks

All Endoscopy lists will adhere to the National Safety Standards for Invasive Procedures (NatSSIPs 2) [15].

5.6 Topical Anaesthesia

- All patients undergoing OGD procedures should be offered topical anaesthesia in the form of a 4% lidocaine spray, unless there is a contraindication to the use of the topical anaesthetic.

5.7 Sedative agents

- The choice of sedative agent is at the discretion of the anaesthetist.
- Of note the ProSed2 trial [5] and others, have shown monotherapy with a Propofol infusion has significant advantages to the use of a combination of sedative drugs. This includes improved patient experience and recovery post procedure with decreased incidence of complications.

5.8 Reversal of sedation or opiates

- Flumazenil can be used to reverse the effects of benzodiazepines and naloxone reverses the effects of opiods. The standard initial adult doses of these drugs are;
    - Flumazenil 200 micrograms iv
    - Naloxone 200 to 400 micrograms iv
- Titrate the appropriate drug to achieve reversal of the relevant agent. Further doses may be required. Consider transfer of the patients involved to main theatre recovery.
- The use of Flumazenil or Naloxone requires reporting via the Datix system when it involves patients having procedural sedation.

5.9 Oxygen Therapy.

- All patients should receive supplemental oxygen during sedation and in recovery. High flow nasal oxygenation (HFNO) may be required for some patients.

5.10 Airway Management

- Patients with known or suspected difficult airways should not be managed in endoscopy and should be booked on a theatre list as per Appendix A
- It is expected that the majority of patients undergoing ACS will maintain their own airways. However, anaesthetists working in endoscopy can manage the airway as they would in a main theatre setting.

- Certain presentations and procedures may have a higher aspiration risk This should be identified pre operatively and if an endotracheal tube is deemed necessary then this should be booked on a theatre list as per Appendix A
- If a supraglottic airway or endotracheal tubes has been inserted, then it must be removed prior to transferring to recovery and the patient must be awake enough to obey commands.
- A standardised difficult airway trolley including a videolaryngoscope, is available in endoscopy. The nearest fibreoptic bronchoscope is in Theatre 8 in the maternity unit.

5.11 Emergencies

- Providing anaesthesia in a remote unit (Endoscopy) is associated with an increased difficulty in obtaining skilled assistance in an emergency. All staff should be aware of working in isolation and how to obtain help in a timely manner.
- Challenging cases, which are felt to be borderline for the endoscopy unit, should be discussed with the on-call consultant anaesthetist and if necessary postponed and rebooked on to a main theatre list.
- In the event of an anaesthetic emergency fast bleep the on-call anaesthetic consultant (300) and the on-call SAS anaesthetist (508) and clearly state 'Endoscopy Unit"
- In the event of a cardiac arrest use the crash bleep system on Ext 2222 and clearly state 'Endoscopy Unit'

5.12 Discharge Criteria

This will follow the standard Trust guidelines for PACU.

https://royaldevonstaff.nhs.uk/search?term=PACU+discharge+criteria&search=Search&searchType=all

- The patient is fully conscious, able to maintain a clear airway and has regained protective airway reflexes.
- Breathing and oxygenation are satisfactory
- The cardiovascular system is stable, with no unexplained cardiac irregularity or persistent bleeding. The specific values of pulse and blood pressure should approximate to normal pre-operative values or be at an acceptable level, ideally within parameters set by the anaesthetist.
- Pain and postoperative nausea and vomiting should be adequately controlled.
- The temperature should be within acceptable limits

- It would be recommended that all patients having sedation should not drive or operate heavy machinery for 24 hours post procedure as per general anaesthesia guidelines

6. MONITORING COMPLIANCE WITH THIS GUIDELINE

6.1 Datix reports will be identified by the Anaesthetic Governance lead and discussed at both the Endoscopy Users group and Anaesthetic Governance meetings.

7. ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

Pre Assessment Guidelines Northern

Peri-operative Guidelines for the Management of Patients with Diabetes

Consent for anaesthesia

PACU Discharge Criteria Guideline

Pre Operative Fasting Guidelines for Elective Surgery

7. PUBLICATION DETAILS

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8. REFERENCES

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(2) https://www.aomrc.org.uk/wp-content/uploads/2021/02/Safe_sedation_practice_for_healthcare_procedures_update_0521.pdf

(3) AG (Joint Advisory Group on GI Endoscopy)
Guide to meeting quality and safety standards
https://www.thejag.org.uk/JAGguidance

(4) Sidhu R, Turnbull D, Newton M, *et al*. *Frontline Gastroenterology 2019;***10**:141–147.

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https://anaesthetists.org/Home/Resources-publications/Guidelines/National-Essential- Anaesthesia-Drugs-List-NEADL

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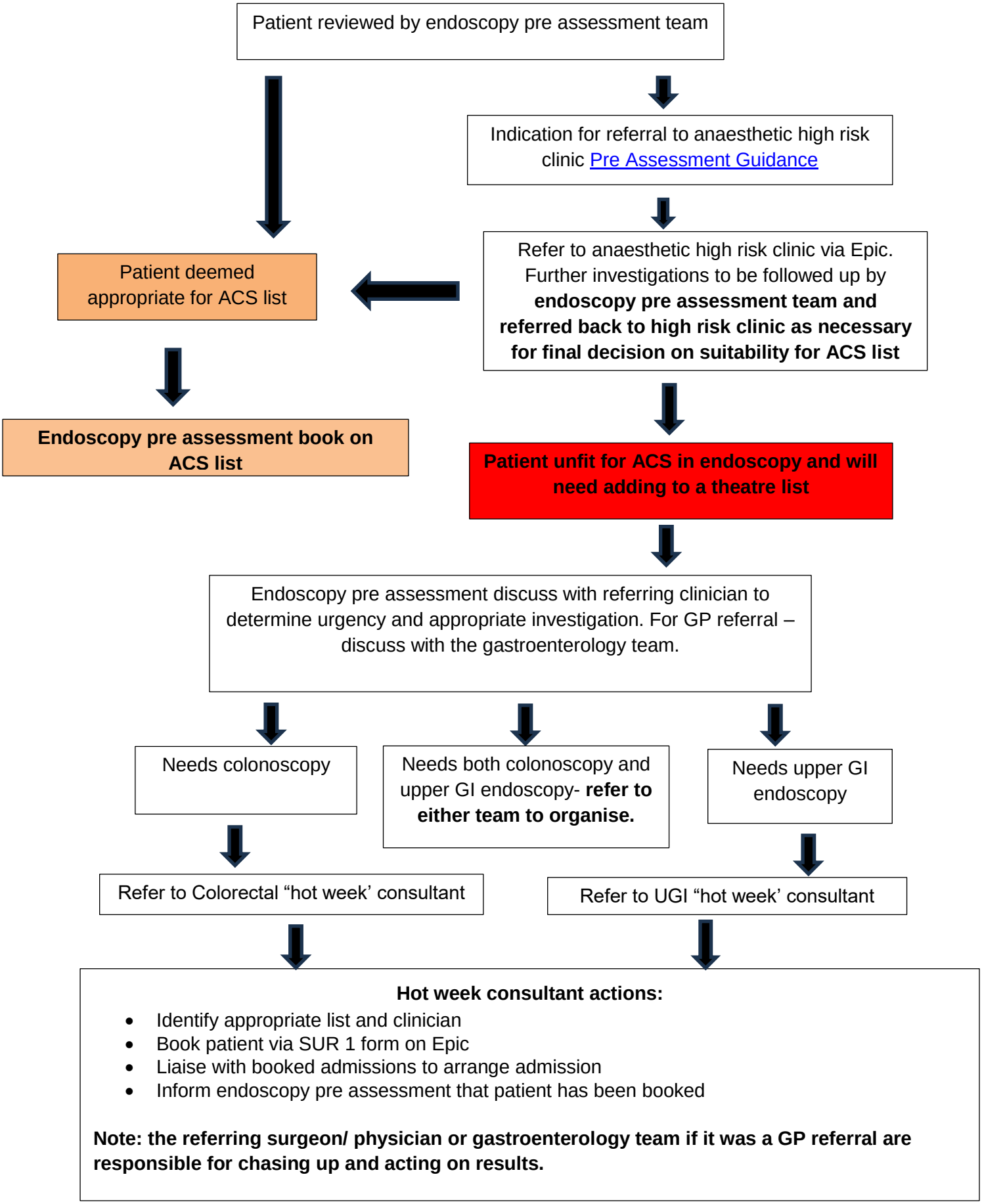
(15) The National Safety Standards for Invasive Procedures 2

https://cpoc.org.uk/guidelines-resources-guidelines/national-safety-standards-invasive-procedures-natssips

(16) Association of Anaesthetists of Great Britain and Ireland. AAGBI: Consent for anaesthesia 2017. *Anaesthesia* 2017; 72: 93-105.

9. APPENDICES

Appendix A: Pathway for Patients Requiring ACS in Endoscopy



Appendix B

Patients presenting with neuromuscular disease

The following guidance from the Motor Neurone Disease Peninsular Network Service defines which patients with MND are suitable for PEG insertion on the Northern site:

- FVC >50% predicted – safe for percutaneous endoscopic gastrostomy (PEG) at NDDH
- FVC 30-50% predicted: Refer for radiologically inserted gastrostomy (RIG) on the Eastern Site. Patient must be able to lie flat for 30 mins with SpO2 >95% and no airway compromise.

Appendix C Essential Drug List for Endoscopy

Emergency Drugs

- Adrenaline mini jets 1/10,000
- Adrenaline 1/1000
- Adenosine
- Amiodarone (mini jet or ampoules)
- Atropine
- Dantrolene
- Ephedrine
- Flumazenil
- Hypostop
- Labetalol
- Magnesium sulphate
- Metaraminol
- Metoprolol
- Naloxone
- Sugammadex
- Suxamethonium

Standard Drugs

- Alfentanil
- Atracurium
- Dexamethasone
- Fentanyl
- Glycopyrrolate
- Glycopyrollate/ neostigmine
- Lidocaine
- Midazolam
- Ondansetron
- Paracetamol
- Propofol
- Rocuronium
- Sevoflurane

Appendix D

Minimal Anaesthesia Equipment Requirements

Airway and Breathing

- Standardised difficult airway trolley including McGrath video laryngoscope and equipment for FONA
- High Flow Nasal Oxygenation device
- Working suction with Yankauer and suction catheters
- Nasal prongs with capnography
- Hudson mask with capnography
- Self-inflating Bag (Ambu-bag)
- Facemasks and HME filters
- Airway adjuncts, including Guedal airway and nasopharyngeal airways.
- Laryngeal mask airway (LMA)/I-Gels sizes 3 to 5
- Laryngoscopes MAC blades 3 & 4
- Bougies including a flexi tipped bougie
- Magills
- Endotracheal tubes (ETT) size 5 to 8
- Tube ties
- Lubricant
- Stylet

Circulation

- Selection of venflons, chlorhexidine wipes, tourniquets and dressings
- Posiflush
- Plasmalyte 1L bags
- Giving sets and 3 way taps
- Blood giving set
- PCA giving sets for TCI
- Pressure bags

Miscellaneous

- Anaesthetic machine and monitor
- Anaesthetic gas scavenging
- Target Controlled infusion pumps x 2 with power supply
- Alternate oxygen supply
- Thermometer
- Nerve stimulator